AROI: Annotated Retinal OCT Images Database

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***Abstract* - The development of optical coherence tomography (OCT) devices has significantly influenced diagnostics and therapy guidance in ophthalmology. The growing number of available images results in the increasing importance of introducing robust algorithms for automatic segmentation in clinical practice. With advances in computer vision in recent years, development of algorithms for segmentation of the retinal structure and/or pathological biomarkers have intensified. However, we are experiencing a reproducibility crisis due to a lack of openly available databases. In this paper we give an overview of a new openly available Annotated Retinal OCT Image (AROI) database that we have developed as a result of the collaboration of one research institution and one hospital. It consists of 1136 annotated B-scans (from 24 patients suffering from age-related macular degeneration) and associated raw high-resolution images. In each B-scan, three retinal layers and three retinal fluids were annotated by an ophthalmologist. Results for intra- and inter-observer errors are obtained to set a baseline for ML algorithms validation. We believe that the AROI database offers many possibilities for the computer vision research community specialized in retinal images and represents a step towards developing a robust artificial intelligence system in ophthalmology.**

Keywords - annotated image database; retinal OCT images; automatic segmentation

I. Introduction

Artificial intelligence and deep learning (DL) algorithms have had many breakthroughs in recent years partially due to large datasets like ImageNet [1]. Given that there are no similar datasets (in terms of diversity, scale, and open access) in medicine, transfer learning is often used. However, it has not been widely adopted in ophthalmology [2]. It is still not clear enough how much transfer learning can help in case where the source dataset consists of natural images (as the ImageNet) and the target dataset consists of medical images (entirely different from natural images) [3]–[5], or how suitable the neural network architectures developed on ImageNet will be for medical images [6].

Since the introduction of optical coherence tomography (OCT), OCT imaging is increasingly present in ophthalmic clinical practice. Despite the lack of publicly available databases, the importance of automatic segmentation has been recognized in the research community. Simultaneously with the progress of DL algorithms in computer vision, in ophthalmology there are increasing requirements for automatic segmentation. We are faced with growing amount of data (a single OCT volume usually consists of over 50 B-scans, depending on the type of OCT device), manual segmentation is time-consuming (and therefore infeasible in clinical practice), and at the same time segmentation of retinal structure and pathological biomarkers are necessary for more accurate diagnosis and more effective therapy. All of the aforementioned issues resulted in a large number of published papers for automatic segmentation in recent years. It is predominantly an outcome of introducing U-net architecture [7] in 2015 which accomplished good results in a small data regime.

The lack of publicly available databases hampers the validation of algorithms and leads to reproducibility/replication and therefore reliability crises. Recent study done by Wintergerst *et al.* [8] attempted to replicate Chen *et al.* algorithm [9] (well-known algorithm for drusen segmentation in OCT images) according to the report in the original paper (as code is not open source). After replication on a different dataset, inferior results were obtained. Even with improvements of the algorithm, originally reported results could not be achieved. To our knowledge it is the first such replication study which raises concerns about reliability/reproducibility. Another problem with algorithms trained on small datasets with lack of diversity is their considerable bias and no possibility of generalization. The reliability of published results is paramount for applying automatic segmentation and AI into clinical practice. Also, the algorithms applied in clinical practice should generalize well, and that could be achieved by training on large and diverse databases (covering all eye diseases as well as healthy persons; with variability in demographic characteristics; different types of OCT devices) – and this can be managed by a public repository with databases collected from various research groups. As labelling is time-consuming and requires expertise, this can be achieved quicker by pooling efforts across research groups.

In this paper we give a detailed overview of a new openly available **A**nnotated **R**etinal **O**CT **I**mage (AROI) database that consists of high-resolution raw OCT images for 24 patients suffering from neovascular age-related macular degeneration (nAMD) with 1136 images (B-scans) annotated. Three retinal layers and three retinal fluids occurring in nAMD have been manually annotated by an ophthalmologist. Inter- and intra-observer errors were calculated. Images are prepared for semantic segmentation and available in PNG format. We describe the current stage of development of the AROI database, with plans for further improvements and new features.

A. Related Databases

The common practice of algorithm validation is through community challenges due to standardized benchmarks (same dataset and same validation metrics). In 2017 “The Retinal OCT fluid Detection and Segmentation Benchmark and Challenge” (RETOUCH) [10] was organized. The benchmarking resource database consisted of OCT volumes from 112 patients suffering from age-related macular degeneration (AMD) and retinal vein occlusion (RVO). OCT volumes were acquired from three types of OCT devices: Topcon, Spectralis, and Cirrus. Three types of retinal fluids were annotated: subretinal fluid (SRF), pigment epithelial detachment (PED), and intraretinal fluid (IRF). [11] To our knowledge, it has been the largest such publicly available database up to date.

In a recent review paper, Khan *et al.* [12] give an extensive overview of publicly available databases in ophthalmology (up to May 2020). According to their study, there are 94 open access datasets. AMD is present in 16% of datasets; 19% of all datasets include images from OCT or OCT angiography; and 35% of datasets include manually annotated segmentations. It leads the authors to raise serious concerns about “data poverty” and argue about challenges in data collection.

An extensive overview of all available databases is beyond the scope of this paper. One of the common problems is that most of the publicly available databases have never become popular due to lack of visibility, accessibility, transparency, absence of data descriptions, etc. However, we would like to highlight the DUKE dataset [13], the most popular used by research groups in developing and evaluating algorithms. It consists of 110 B-scans of 10 DME (diabetic macular edema) patients acquired with Spectralis OCT device.

II. PROPERTIES OF AROI DATABASE

A. Data Collection

Age-related macular degeneration (AMD) is the leading cause of irreversible blindness in people 50 years of age or older in the developed world [14]. The total lifetime prevalence in the world is 8.69% [15] and it is likely to increase due to growing ageing of population. According to these estimates, by 2040 there will be 288 million people affected by AMD [15].

AMD is a progressive disease. The early and intermediate stage is usually asymptomatic, and the only known therapy, for now, are antioxidant and mineral supplements that can eventually slow the progress. Late stage is divided into exudative, wet or neovascular AMD (nAMD) and geographic atrophy (GA). In the case of neovascular AMD (Figure 1), the mainstay of treatment are intravitreal injections of anti-Vascular Endothelial Growth Factor (anti-VEGF) drugs. In the case of geographic atrophy, characterized by geographic areas of retinal pigment epithelium and photoreceptor loss, there is no adequate therapy for now [16].

Diagnosing and monitoring AMD is significantly enhanced by introducing optical coherence tomography (OCT). OCT is a method of medical imaging which enables non-invasive 3D imaging. 3D retinal images of high resolution enable clinical evaluation of the retinal disease as morphological features can be observed and measured from OCT images. It has significantly improved the diagnosis and treatment of AMD. For the AROI database, macular SD-OCT volumes were recorded with the Zeiss Cirrus HD OCT 4000 device. Each OCT volume (Figure 2) consists of 128 B-scans (2D images in XZ plane) with a resolution of 1024 x 512 pixels with pixel size 1.96 x 11.74 μm. Although terms like “OCT volume” and “3D imaging” are common, more precisely the volume consists of 128 B-scans spaced 46.78 μm apart.

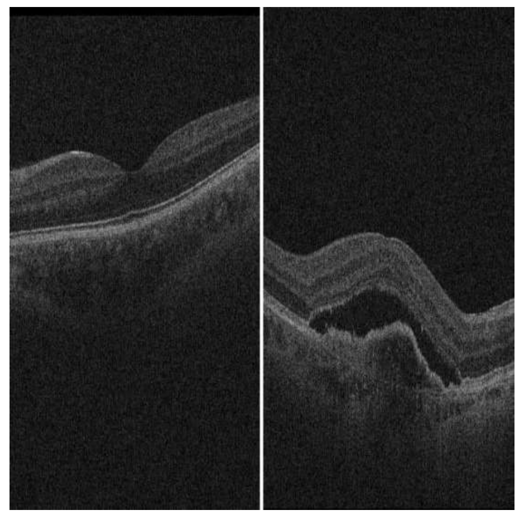

Figure 1. Example of a B-scan (in the fovea, the central part of macula) for the healthy person (left) and patient suffering from nAMD (right).

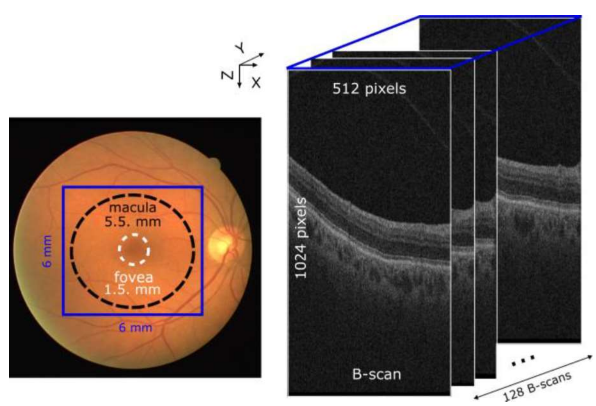

Figure 2. Fundus image and macular OCT volume.

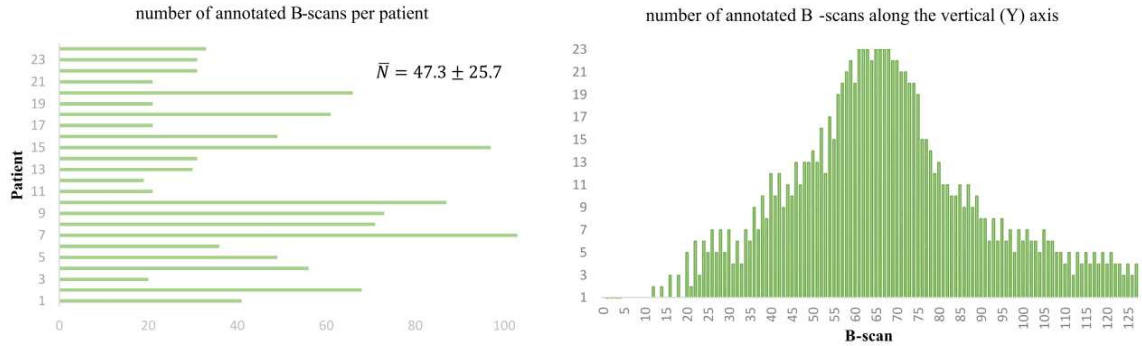

Figure 3. The number of annotated B-scans for each individual patient (left) and the distribution of the number of annotated B-scans along vertical (Y) axis (right).

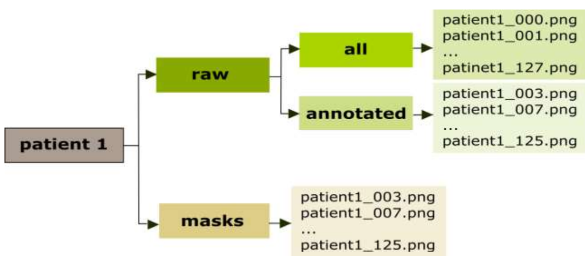

Figure 4. Organization of images in the database.

We have collected and annotated OCT images for 24 patients diagnosed with nAMD undergoing active anti-VEGF therapy or follow-up at a tertiary hospital between April 1, 2018, and October 30, 2018. Data collection adhered to the tenets of the Declaration of Helsinki and the standards of Good Scientific Practice of Sestre milosrdnice University Hospital Center (Zagreb, Croatia). All patients signed informed consent and the data were anonymized. The presented study was approved by the Ethics Committee of the Sestre milosrdnice University Hospital Center (EP-3272/18-11) and the Faculty of Electrical Engineering and Computing (Zagreb, Croatia).

All annotations were done by an ophthalmologist in GNU Image Manipulation Program (GIMP) [17]. The database consists of 1136 annotated B-scans out of total of 3072 B-scans for 24 patients (37% of B-scans were annotated). Not every single B-scan would be annotated if adjacent scans were similar. Also, in the fovea (the central part of macula), a larger number of B-scans were annotated as visual acuity mostly depends on the pathological changes in that area. Figure 3. shows the number of annotated B-scans for each individual patient. The average number of annotated B-scans per patient is 47.3 (with standard deviation of 25.7). The distribution of the number of annotated B-scans along vertical (Y) axis is as well shown in Figure 3. It is seen that the major number of annotated scans is around the foveal center (ordinal number 62).

Images from OCT devices could be exported in IMG format, and in the database, we have made them available in more common and accessible PNG format. Also, images are grouped so that for each patient there are all raw B-scans, with annotated ones provided separately. The name of each image is associated with the patient ordinal number (1-24) and B-scan ordinal number (0-127) (Figure 4). B-scans which are not annotated are also available as adjacent scans provide contextual information that can be used in case of developing a 3D segmentation method. At this stage, the images are not divided into training, validation, and test set – division could be optimized depending on the developing method. However, we do not recommend splitting B-scans from the same patient over training, validation, and test set as adjacent B-scans are similar, and therefore in the test set there will be "already seen" B-scans in the training set (causing overestimated evaluation of the method). We recommend k-fold cross-validation where folds are formed such that they do not split images from same patients into different folds.

B. *Annotations of retinal layers and fluids*

Four boundaries between the layers are annotated (Figure 5). The retina is a layered tissue inside the eye and is histologically divided into ten layers: 1. internal limiting membrane (ILM), 2. retinal nerve fiber layer (RNFL), 3. ganglion cell layer (GCL), 4. inner plexiform layer (IPL), 5. inner nuclear layer (INL), 6. outer plexiform layer (OPL), 7. outer nuclear layer (ONL), 8. external or outer limiting membrane (ELM or OLM), 9. photoreceptor layer and 10. retinal pigment epithelium/Bruch membrane complex (RPE-BM). The layers seen in the OCT scan correspond to these histological layers. As pathological changes can lead to large deviations in the structure of the layers, it is often impossible for an ophthalmologist reading the OCT scan to determine all 11 boundaries or 10 layers reliably. We selected boundaries that can be readily determined in virtually all images and that are also relevant for localizing the observed fluids: internal limiting membrane – ILM, retinal pigment epithelium (RPE), and Bruch membrane (BM). Additionally, the boundary between the inner plexiform layer and inner nuclear layer (IPL/INL) was annotated as the foveal center could be defined as the point of least distance between ILM and IPL/INL.

Of the fluids, the following are annotated (Figure 6): pigment epithelial detachment (PED), subretinal fluid and subretinal hyperreflective material (marked jointly as SRF), and intraretinal fluid (IRF).

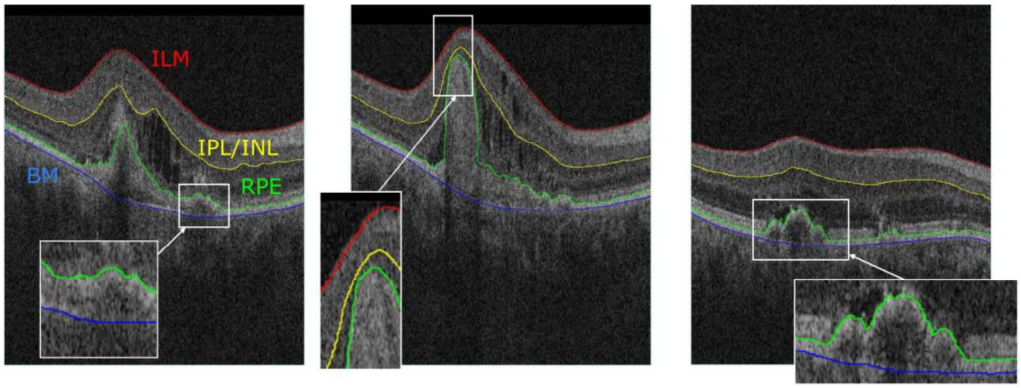

Figure 5. Examples of annotated boundaries between layers: internal limiting membrane – ILM, boundary between the inner plexiform layer and inner nuclear layer (IPL/INL), retinal pigment epithelium (RPE), and Bruch membrane (BM). Images are cropped and only the ROI is visible.

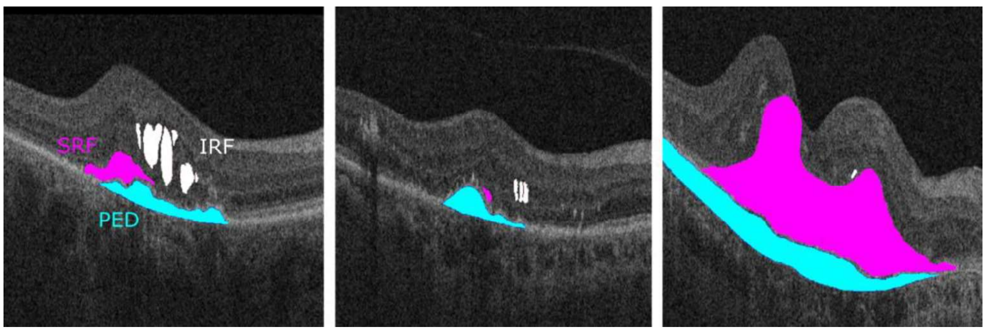

Figure 6. Examples of annotated fluids: pigment epithelial detachment (PED), subretinal fluid (SRF), and intraretinal fluid (IRF). Images are cropped and only the ROI is visible.

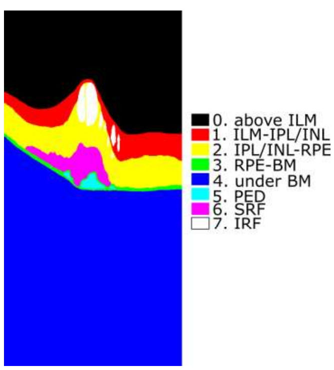

Figure 7. Example of an image prepared for semantic segmentation (with eight classes).

C. Images prepared for semantic segmentation

Joint segmentation of layers and fluids should lead to better accuracy due to the dependence of layers and fluids. To determine only the annotated boundaries, it is more appropriate to make semantic segmentation first and subsequently to determine the boundaries. For that reason images are prepared for semantic segmentation (Figure 7) with eight classes: area above the retina (vitreous), three retinal surfaces (we use the term "surface" to label the area between annotated boundaries and to distinguish them from 10 retinal layers since they do not match; surface usually covers more layers), area below the retina (choroid) and three retinal fluids (PED, SRF, IRF).

A significant class imbalance is one of the reasons that make automatic segmentation challenging and should be considered when developing the algorithm for automatic segmentation. Shares for each class in the total number of pixels are shown in Table I. The background (area above ILM and under BM) occupies as much as 83.26% while the IRF occupies only 0.12% in the total number of pixels. The higher segmentation errors are expected for classes that are less represented. Out of a total of 1136 scans, PED, SRF, and IRF are present in 1014 (89.26%), 648 (57.04%), and 229 (20.16%) B-scans, respectively.

TABLE I. PERCENTAGE OF THE NUMBER OF PIXELS FOR EACH CLASS IN THE TOTAL NUMBER OF PIXELS

Above ILM	ILM - IPL/INL	IPL/INL - RPE	RPE - BM	Under BM	PED	SRF	IRF
35.07%	5.10%	7.89%	1.07%	48.19%	1.50%	1.05%	0.12%

TABLE II. INTRA- AND INTER-OBSERVER ERROR IN BOUNDARIES ASSESSMENT

	Inter-observer error		Intra-observer error	
	MSE	MAE	MSE	MAE
ILM	5.87±3.68	1.86±0.64	2.10±1.43	1.02±0.36
IPL/INL	20.46±47.71	2.65±1.19	5.93±4.80	1.70±0.59
RPE	51.74±148.15	2.96±2.30	8.28±9.33	1.80±0.79
BM	12.77±17.79	2.54±1.59	5.17±6.68	1.53±0.78

TABLE III. THE DICE SCORE (MEAN AND STANDARD DEVIATION) IN THE INTER-OBSERVER AND THE INTRA-OBSERVER CASE.

	Above ILM	ILM - IPL/INL	IPL/INL - RPE	RPE - BM	Under BM	PED	SRF	IRF
Inter-observer	0.982 ±0.072	0.950 ±0.111	0.948 ±0.112	0.699 ±0.129	0.989 ±0.114	0.860 ±0.301	0.876 ±0.366	0.735 ±0.280
Intra-observer	0.998 ±0.003	0.973 ±0.008	0.970 ±0.117	0.778 ±0.092	0.998 ±0.001	0.912 ±0.242	0.924 ±0.331	0.844 ±0.140

D. Inter- and intra-observer error

The same ophthalmologist re-annotated 75 randomly chosen B-scans with no reference to the original ones and with time delay (3 months after finishing the first annotations) to determine the intra-observer error. Also, the same 75 B-scans were annotated by an additional ophthalmologist to estimate inter-observer error thus enabling the validation of developed algorithms in relation to human variability.

To determine the intra-observer error, we have calculated the differences between "ground truth" (original annotations from 1st ophthalmologist) and re-annotations from the same ophthalmologist. To determine the inter-observer error, we have calculated the differences between "ground truth" and annotations from 2nd ophthalmologist. Variations in assessment of retinal boundaries are presented with mean square error (MSE) and mean absolute error (MAE) with belonging standard deviations (Table II). Values are shown in pixels where each pixel corresponds to 1.96 µm along the axial (Z) axis.

The most common metric for estimating error used in semantic segmentation is the Dice score. The Dice score ranges from 0 to 1, where 1 indicates perfect matching. In Table III the Dice scores for all eight classes are presented in case of evaluating inter- and intra-observer error. It can be seen that the values are the lowest for IRF and the area between RPE and BM which are represented by only 0.12% and 1.07% pixels, respectively (Table I).

Considering the lack of standardization in measuring inter- and intra-observer variability in ophthalmology there are a lot of rising questions about how to properly measure inter- and intra-observer variability (how to choose the size of the dataset, should images be chosen randomly, or to best present the whole dataset, how many observers to choose, etc.) According to [18] it would be better to choose three annotators. In our opinion it would be preferable if they are not from the same hospital (the greater probability of the introducing same bias) – as it was not manageable in this study, we plan further investigation in future work.

III. DISCUSSION

In this paper, we have presented a new publicly available AROI database – database of annotated retinal OCT images for 24 patients with nAMD. As images are prepared for semantic segmentation, exported in PNG format, and organized within the database, the database is accessible and easy to use within the broader research community. Also, intra- and inter-observer errors were calculated which provides insight into the estimation of human error.

Training ML algorithms on databases that are not diverse enough would lead to considerable bias which is unacceptable in clinical practice. The AROI database is limited in scale and diversity but our goal was not building a broad and comprehensive database as that is not the feasible task for a single research group. We have limited ourselves to a narrow segment of the population, one type of disease, and one type of OCT device, but at the same time we have made significant efforts to secure the quality of annotations.

The main challenge to the introduction of automatic segmentation and, more generally, AI into clinical practice is not in developing more accurate architectures of neural networks but in overcoming the lack of well-annotated open access data. We believe that the AROI database represents a step towards developing a robust artificial intelligence system in ophthalmology.

OPENLY AVAILABLE DATASET

The image database is openly available to other research group members and can be accessed by following the instructions provided in [19] and through the Open Science Framework repository [20].

ACKNOWLEDGMENT

Special thanks to Aida Kasumović, MD (Department of Ophthalmology, Sestre milosrdnice University Hospital Center, Zagreb, Croatia) who did image labeling for the purpose of calculating inter-observer error. Also, authors would like to thank Filip Melinščak for valuable discussions and comments on the text.

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